

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

np.random.seed(42)
```

```
n = 1000

data = pd.DataFrame({
    'Student_ID': range(1, n+1),

    'Source': np.random.choice(
        ['LinkedIn', 'Instagram', 'Website', 'Referral'],
        n),

    'Course': np.random.choice(
        ['Data Analytics',
         'Web Development',
         'Python',
         'AI/ML'],
        n),

    'Gender': np.random.choice(
        ['Male', 'Female'],
        n),

    'Applied': np.random.choice(
        ['Yes', 'No'],
        n,
        p=[0.8, 0.2])
})
```

```
data['Assessment'] = np.where(
    data['Applied']=='Yes',
    np.random.choice(['Yes', 'No'],
                     n,
                     p=[0.75,0.25]),
    'No'
)

data['Interview'] = np.where(
    data['Assessment']=='Yes',
    np.random.choice(['Yes', 'No'],
                     n,
                     p=[0.7,0.3]),
    'No'
)

data['Selected'] = np.where(
    data['Interview']=='Yes',
    np.random.choice(['Yes', 'No'],
                     n,
                     p=[0.65,0.35]),
    'No'
)

data['Completed'] = np.where(
    data['Selected']=='Yes',
    np.random.choice(['Yes', 'No'],
                     n,
                     p=[0.8,0.2]),
    'No'
)

data.head()
```

	Student_ID	Source	Course	Gender	Applied	Assessment	Interview	Selected	Completed
0	1	Website	Web Development	Male	Yes	No	No	No	No
1	2	Referral	Python	Female	Yes	No	No	No	No
2	3	LinkedIn	Data Analytics	Female	Yes	Yes	No	No	No
3	4	Website	Data Analytics	Female	Yes	No	No	No	No
4	5	Website	Data Analytics	Female	Yes	Yes	No	No	No

```
print(data.shape)

print(data.info())
```

```
print(data.describe(include='object'))
```

```
(1000, 9)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 9 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   Student_ID  1000 non-null   int64
1   Source       1000 non-null   object
2   Course       1000 non-null   object
3   Gender       1000 non-null   object
4   Applied      1000 non-null   object
5   Assessment   1000 non-null   object
6   Interview    1000 non-null   object
7   Selected     1000 non-null   object
8   Completed    1000 non-null   object
dtypes: int64(1), object(8)
memory usage: 70.4+ KB
None
```

	Source	Course	Gender	Applied	Assessment	Interview	Selected	\
count	1000	1000	1000	1000	1000	1000	1000	
unique	4	4	2	2	2	2	2	
top	Referral	Data Analytics	Male	Yes	Yes	No	No	
freq	280	267	501	807	600	561	720	

	Completed
count	1000
unique	2
top	No
freq	785

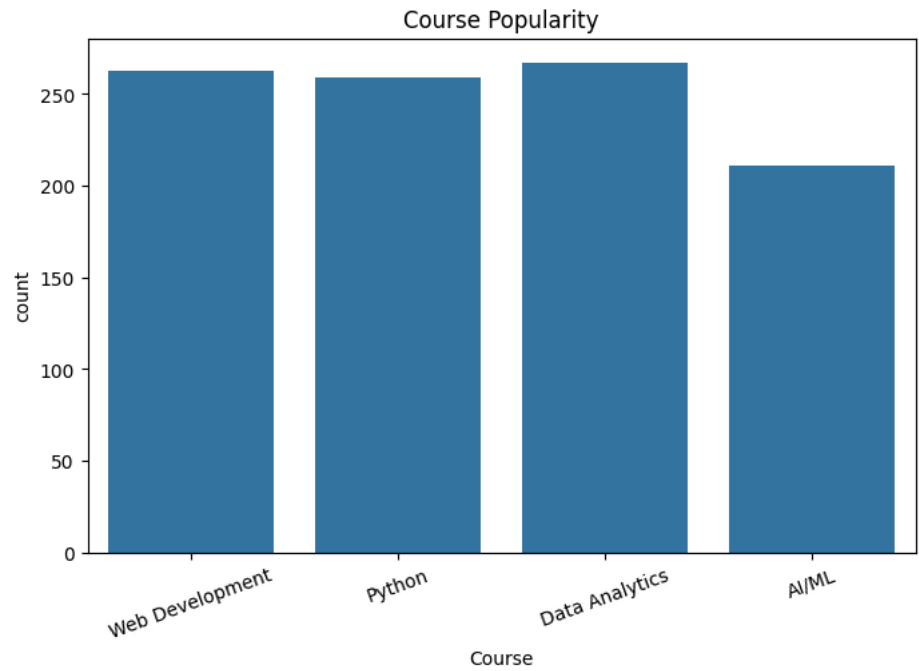
```
print(data.isnull().sum())
```

```
Student_ID    0
Source        0
Course        0
Gender        0
Applied       0
Assessment    0
Interview     0
Selected      0
Completed     0
dtype: int64
```

```
plt.figure(figsize=(8,5))

sns.countplot(
    x='Course',
    data=data)

plt.title("Course Popularity")
plt.xticks(rotation=20)
plt.show()
```

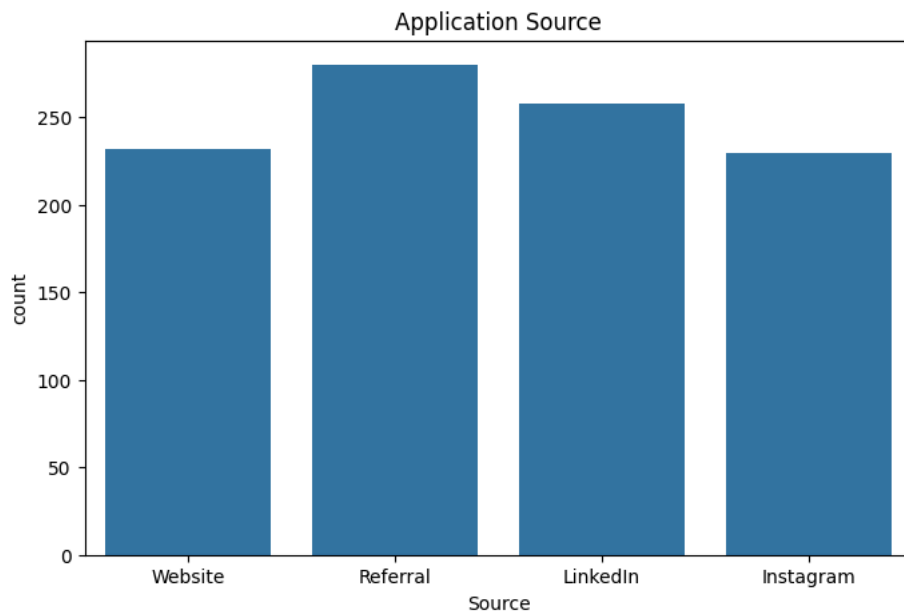


```
plt.figure(figsize=(8,5))

sns.countplot(
    x='Source',
```

```
data=data)

plt.title("Application Source")
plt.show()
```

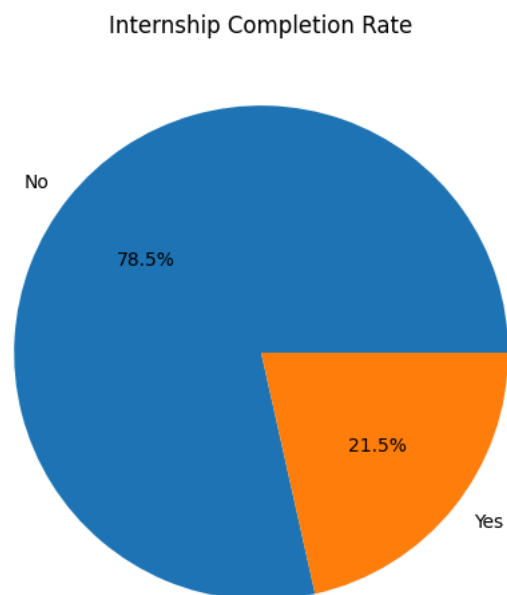


```
completion = data['Completed'].value_counts()

plt.figure(figsize=(6,6))

plt.pie(
    completion,
    labels=completion.index,
    autopct='%1.1f%%')

plt.title("Internship Completion Rate")
plt.show()
```



```
funnel = [
    len(data),

    len(data[data['Applied']=='Yes']),

    len(data[data['Assessment']=='Yes']),

    len(data[data['Interview']=='Yes']),

    len(data[data['Selected']=='Yes']),

    len(data[data['Completed']=='Yes'])
]
```

```

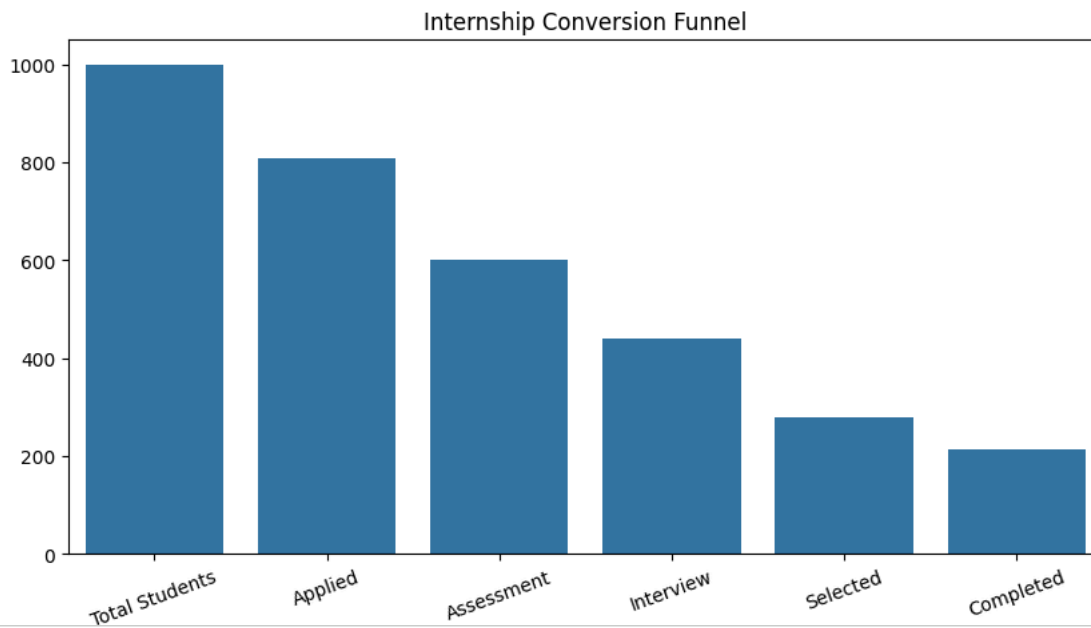
stages = [
    'Total Students',
    'Applied',
    'Assessment',
    'Interview',
    'Selected',
    'Completed'
]

plt.figure(figsize=(10,5))

sns.barplot(
    x=stages,
    y=funnel)

plt.title("Internship Conversion Funnel")
plt.xticks(rotation=20)
plt.show()

```



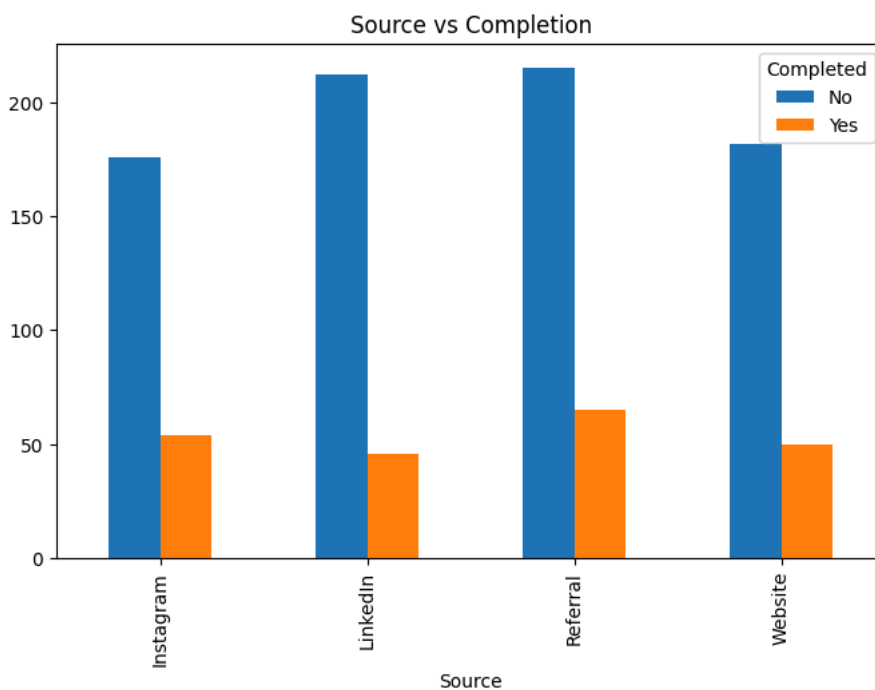
```

source_completion = pd.crosstab(
    data['Source'],
    data['Completed'])

source_completion.plot(
    kind='bar',
    figsize=(8,5))

plt.title("Source vs Completion")
plt.show()

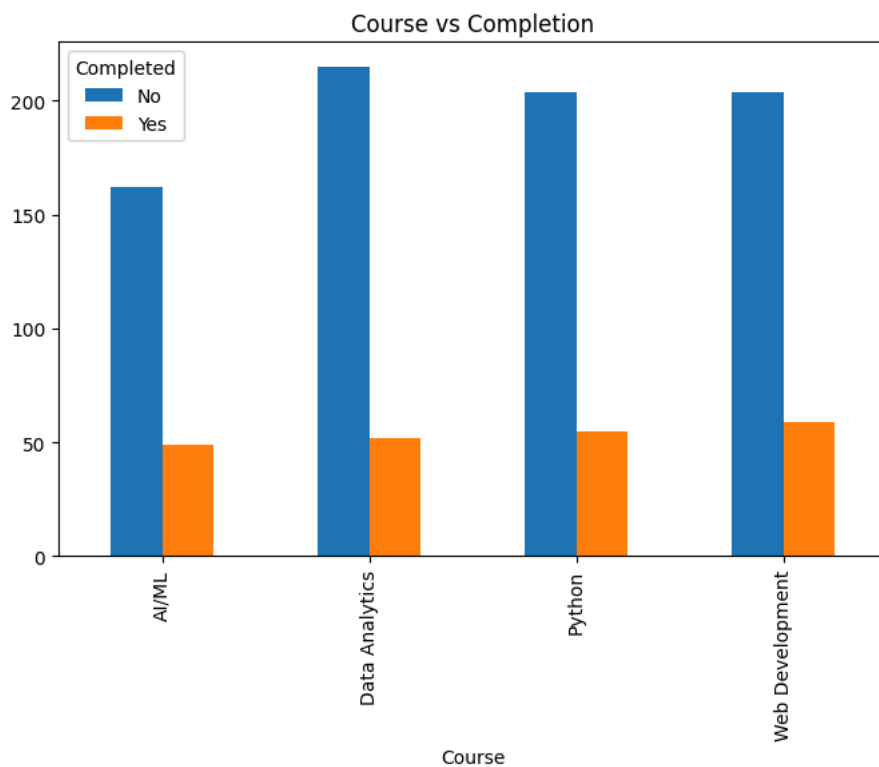
```



```
course_completion = pd.crosstab(
    data['Course'],
    data['Completed'])
```

```
course_completion.plot(
    kind='bar',
    figsize=(8,5))
```

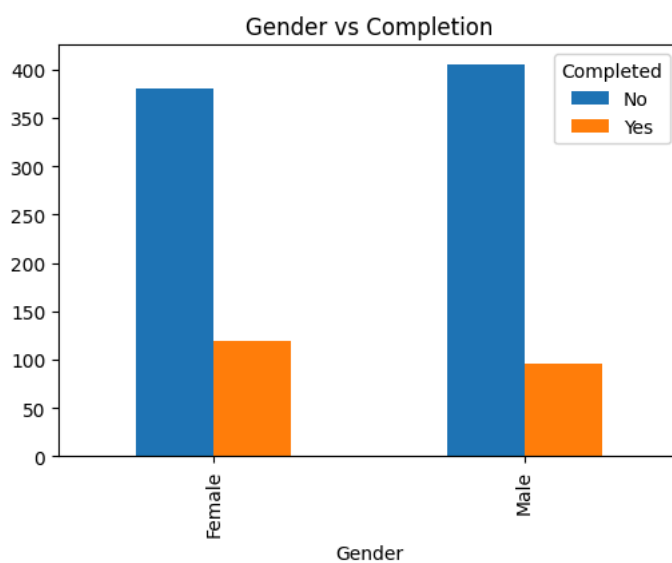
```
plt.title("Course vs Completion")
plt.show()
```



```
gender_completion = pd.crosstab(
    data['Gender'],
    data['Completed'])
```

```
gender_completion.plot(
    kind='bar',
    figsize=(6,4))
```

```
plt.title("Gender vs Completion")
plt.show()
```



```
print("Total Students :", len(data))
print("Completed Internship :",
      len(data[data['Completed']=='Yes']))
```

```
Total Students : 1000
Completed Internship : 215
```

